

Experiment- 8

P-N-Diode

AIM:

To draw the V-I characteristic curve of a p-n junction in forward bias and reverse bias.

APPARATUS:

p-n-junction diode kit.

THEORY:

Forward bias: A p-n junction diode is forward biased when the p side of the diode is connected to the positive terminal of the battery and the n side to the negative terminal. In forward bias, the p-side is at a higher potential with respect to the n-side. When the bias voltage is increased, the forward current increases slowly in the beginning and then rapidly. The voltage at which the forward current shows a sudden increase is known as the knee voltage.

Reverse bias: When the n-region of a p-n junction diode is at a higher potential with respect to the p-region, it is said to be reverse biased. In reverse bias, the p-side of the p-n junction diode is connected to the negative of the battery.

PROCEDURE:

1. Make the connections as shown in the figure.
2. Always connect the positive terminal of the voltmeter and ammeter to the positive terminal of the battery.
3. Now move the contact towards the positive to increase the forward bias voltage slowly; the current remains zero.
4. Increase the forward bias voltage in steps. The current will still be zero. (This is due to the barrier of the junction potential)
5. Increase V_f in steps of 0.2V and note the corresponding current I_f .
6. At a certain value of V_f , the current increases suddenly, and this represents the forward breakdown stage.
7. Record your observations in the table.
8. Plot the graph showing the variation of forward current with forward voltage and reverse current with reverse voltage from the graph

RESULT:

The V-I characteristic curve of a p-n junction diode in forward bias and reverse bias is drawn.

Precautions

- The connections should be neat, clean, and tight.
- The key should be used in the circuit and opened when the circuit is not being used.

Sources of Error

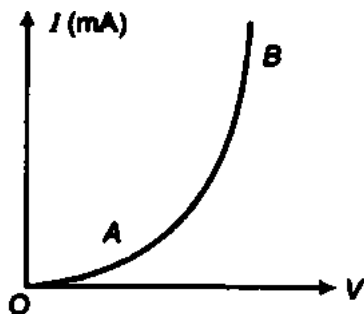
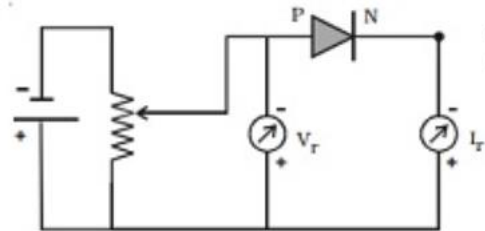
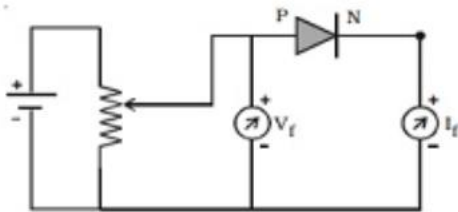
- The junction diode supplied may be faulty.
- The supply may be beyond the tolerance range.

Diagram and observation table on the left side of the page (diagram on one page and observation tables on the other page)

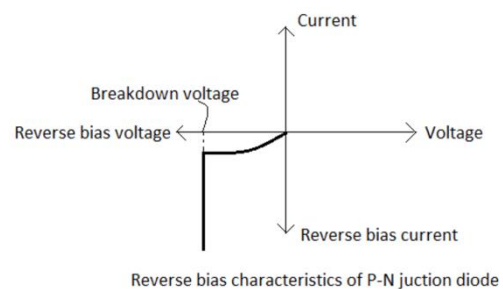
Diagram:

Forward bias

Reverse bias



P-N junction forward bias characteristic curve



Reverse bias characteristics of P-N junction diode

Observation table

Forward bias

Least count of voltmeter=

Least count of ammeter=

S.No.	Forward bias voltage (V_f)	Forward bias current (I_f)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Reverse bias

Least count of voltmeter=

Least count of ammeter=

S.No	Reverse bias voltage (V_r)	Reverse bias current (I_r)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		